

Swagger/OpenAPI Implementation Summary

Overview

Interactive API documentation has been implemented using Swagger/OpenAPI to provide a user-friendly interface for external system developers to explore, test, and integrate with the ISM Sponsor REST API.

Implementation Date: April 8, 2026

Package Version: Swashbuckle.AspNetCore 6.6.2

API Version: v1

What Was Implemented

1. Package Dependencies

File: ISMSponsor.csproj

Added Swashbuckle.AspNetCore package:

```
<PackageReference Include="Swashbuckle.AspNetCore" Version="6.6.2" />
```

Enabled XML documentation generation for API comments:

```
<GenerateDocumentationFile>true</GenerateDocumentationFile>  
<NoWarn>$(NoWarn);CS1591</NoWarn>
```

2. Program.cs Configuration

File: Program.cs

Service Registration (Lines 251-299)

Added Swagger services with comprehensive OpenAPI configuration:

```
builder.Services.AddEndpointsApiExplorer();  
builder.Services.AddSwaggerGen(options =>  
{  
    options.SwaggerDoc("v1", new Microsoft.OpenApi.Models.OpenApiInfo  
    {  
        Title = "ISM Sponsor API",  
        Version = "v1",  
        Description = "REST API for Student Charging Portal, PowerSchool,  
NetSuite, and OBS integration",  
    });  
});
```

```

        Contact = new Microsoft.OpenApi.Models.OpenApiContact
        {
            Name = "ISM Development Team",
            Email = "support@ismsponsor.edu"
        }
    });

    // Include XML documentation comments
    var xmlFile = $"
{System.Reflection.Assembly.GetExecutingAssembly().GetName().Name}.xml";
    var xmlPath = System.IO.Path.Combine(AppContext.BaseDirectory,
xmlFile);
    if (System.IO.File.Exists(xmlPath))
    {
        options.IncludeXmlComments(xmlPath);
    }

    // Configure JWT Bearer authentication for Swagger
    options.AddSecurityDefinition("Bearer", new
Microsoft.OpenApi.Models.OpenApiSecurityScheme
    {
        Description = "JWT Authorization header using the Bearer scheme.
Enter 'Bearer' [space] and then your token.",
        Name = "Authorization",
        In = Microsoft.OpenApi.Models.ParameterLocation.Header,
        Type = Microsoft.OpenApi.Models.SecuritySchemeType.ApiKey,
        Scheme = "Bearer"
    });

    options.AddSecurityRequirement(new
Microsoft.OpenApi.Models.OpenApiSecurityRequirement
    {
        {
            new Microsoft.OpenApi.Models.OpenApiSecurityScheme
            {
                Reference = new Microsoft.OpenApi.Models.OpenApiReference
                {
                    Type =
Microsoft.OpenApi.Models.ReferenceType.SecurityScheme,
                    Id = "Bearer"
                }
            },
            Array.Empty<string>()
        }
    });
});

```

Middleware Configuration (Lines 310-320)

Enabled Swagger UI in Development and Pilot environments:

```
if (app.Environment.IsDevelopment() ||
app.Environment.IsEnvironment("Pilot"))
{
    app.UseSwagger();
    app.UseSwaggerUI(options =>
    {
        options.SwaggerEndpoint("/swagger/v1/swagger.json", "ISM Sponsor
API v1");
        options.RoutePrefix = "api/docs"; // Access at /api/docs
    });
}
```

3. XML Documentation Enhancements

File: `Controllers/Api/CoverageController.cs`

Fixed missing parameter documentation:

- Added `<param name="studentId">` tag
- Added `<param name="from">` tag
- Added `<param name="to">` tag

All other API controllers (`IntegrationController`, `ReferenceController`, `AuditApiController`, `StatementsController`, `SponsorsApiController`) already had comprehensive XML documentation from previous implementation.

How to Access Swagger UI

Development Environment

1. Start the application:

```
dotnet run
```

2. Navigate to Swagger UI:

```
https://localhost:5001/api/docs
```

or

```
http://localhost:5000/api/docs
```

Pilot Environment

1. Deploy to Azure App Service (Pilot slot)
2. Access Swagger UI at:

```
https://ismsponsor-pilot.azurewebsites.net/api/docs
```

Production Environment

Swagger is intentionally disabled in Production for security reasons. External consumers should use the exported OpenAPI specification (see below).

Using Swagger UI

1. Explore API Endpoints

- Browse all available endpoints organized by controller
- View request/response schemas with examples
- See HTTP methods, status codes, and authentication requirements

2. Test API Endpoints

Without Authentication (Public Endpoints):

1. Expand the endpoint you want to test
2. Click "Try it out"
3. Fill in required parameters
4. Click "Execute"
5. View the response

With Authentication (Protected Endpoints):

1. Click the "Authorize" button at the top
2. Enter your JWT token in the format: **Bearer <your-token-here>**
3. Click "Authorize"
4. Now you can test protected endpoints

3. Export OpenAPI Specification

Download the raw OpenAPI JSON specification for use with code generation tools:

```
https://localhost:5001/swagger/v1/swagger.json
```

API Documentation Coverage

Documented Endpoints (25+ endpoints across 6 controllers)

CoverageController (/api/v1/coverage)

-  **POST /evaluate** - Evaluate coverage for a charge line
-  **POST /commit** - Commit coverage decision with audit trail
-  **POST /preview** - Preview coverage without persisting
-  **GET /decisions** - Query coverage decisions (with filters)
-  **GET /decisions/{id}** - Get decision by audit ID
-  **GET /reasons** - Get all reason codes

IntegrationController (/api/v1/integrations)

-  **POST /powerschool/student-sponsor-sync** - Sync student-sponsor links from PowerSchool
-  **POST /netsuite/allocation-post** - Post coverage allocation to NetSuite
-  **POST /obs/statement-update** - Update statement batch in OBS
-  **GET /sync-status** - Get integration sync status by correlation ID

ReferenceController (/api/v1/reference)

-  **GET /sponsors** - Get active sponsors (with search/pagination)
-  **GET /items** - Get billable items by school year
-  **GET /categories** - Get item categories
-  **GET /active-sponsor-links** - Get active sponsor-student links

AuditApiController (/api/v1/audit)

-  **GET /decisions/{decisionId}** - Get coverage decision by audit ID
-  **GET /integrations/{correlationId}** - Get integration sync history

StatementsController (/api/v1/statements)

-  **GET /students/{studentId}** - Get statement data for student
-  **GET /sponsors/{sponsorId}** - Get statement data for sponsor

SponsorsApiController (/api/v1/sponsors)

-  **GET /sponsors/{sponsorId}** - Get detailed sponsor information

Authentication in Swagger

JWT Bearer Token Format

All protected endpoints require a JWT Bearer token in the Authorization header:

```
Authorization: Bearer <your-jwt-token>
```

How to Get a Token

1. **Development/Testing:** Use the application's login page to authenticate and extract the token from the browser's developer tools (Application > Cookies)
2. **Automated Integration:** Implement OAuth 2.0 client credentials flow or service account authentication (future enhancement)

Authorization Matrix

Endpoint	Roles Required
/api/v1/coverage/*	admin, admissions, cashier
/api/v1/integrations/*	admin, cashier
/api/v1/reference/*	admin, admissions, cashier, sponsor
/api/v1/audit/*	admin, cashier
/api/v1/statements/*	admin, cashier, sponsor
/api/v1/sponsors/*	admin, admissions, cashier, sponsor

Code Generation from OpenAPI

Generate C# Client

```
# Install NSwag.MSBuild
dotnet add package NSwag.MSBuild

# Generate client
nswag openapi2csclient \
  /input:https://localhost:5001/swagger/v1/swagger.json \
  /output:ISMSponsorApiClient.cs \
  /namespace:ISMSponsor.Client
```

Generate TypeScript Client

```
# Install openapi-typescript-codegen
npm install -g openapi-typescript-codegen

# Generate client
openapi --input https://localhost:5001/swagger/v1/swagger.json \
  --output ./src/api \
  --client axios
```

Generate Python Client

```
# Install openapi-generator-cli
npm install -g @openapitools/openapi-generator-cli

# Generate client
openapi-generator-cli generate \
  -i https://localhost:5001/swagger/v1/swagger.json \
  -g python \
  -o ./ism_sponsor_client
```

Troubleshooting

Issue: Swagger UI Not Loading

Symptoms: 404 error when accessing `/api/docs`

Solution:

1. Verify you're in Development or Pilot environment (Production disables Swagger)
2. Check that the application is running
3. Verify the URL includes the correct port (5001 for HTTPS, 5000 for HTTP)

Issue: "Authorize" Button Not Working

Symptoms: Authentication fails after entering token

Solution:

1. Ensure token is in the format: `Bearer <token>` (with the "Bearer" prefix)
2. Verify token is not expired
3. Check that the token has the correct roles for the endpoint you're testing

Issue: XML Documentation Not Appearing

Symptoms: Endpoint descriptions are missing in Swagger UI

Solution:

1. Verify XML documentation file exists: `bin/Debug/net8.0/ISMSponsor.xml`
2. Rebuild the project: `dotnet build`
3. Check that `<GenerateDocumentationFile>true</GenerateDocumentationFile>` is in `.csproj`

Issue: 401 Unauthorized on All Endpoints

Symptoms: All endpoints return 401 even with valid token

Solution:

1. Click the "Authorize" button at the top of Swagger UI (not on individual endpoints)
2. Enter your token: `Bearer <your-jwt-token>`

3. Click "Authorize" and "Close"
4. Try the endpoint again

Security Considerations

1. Environment-Specific Access

- **Development:** Swagger UI enabled for local testing
- **Pilot:** Swagger UI enabled for UAT and integration testing
- **Production:** Swagger UI **disabled** - use exported OpenAPI spec only

2. Rate Limiting (Future Enhancement)

Consider implementing rate limiting for Swagger UI endpoints in Pilot to prevent abuse:

```
// Future enhancement: Add rate limiting
builder.Services.AddRateLimiter(options =>
{
    options.AddPolicy("swagger", context =>
RateLimitPartition.GetFixedWindowLimiter(
    partitionKey: context.Connection.RemoteIpAddress?.ToString(),
    factory: _ => new FixedWindowRateLimiterOptions
    {
        Window = TimeSpan.FromMinutes(1),
        PermitLimit = 30
    }));
});
```

3. IP Whitelisting for Pilot

Configure Azure App Service to restrict Swagger UI access to specific IP ranges in Pilot environment.

Build Status

Build Output

```
Build succeeded with 2 warnings in 2.52s
```

Warnings (Acceptable)

```
Program.cs(39,22): warning ASP0000: Calling 'BuildServiceProvider' from application code results in an additional copy of singleton services being created.
```


Status: Acceptable - Required for configuration validation at startup.

XML Documentation

```
bin/Debug/net8.0/ISMSponsor.xml (118 KB)
```

Status: Generated successfully with complete API coverage.

Future Enhancements

1. API Versioning in Swagger

Support multiple API versions (v1, v2) in Swagger UI:

```
options.SwaggerDoc("v1", new OpenApiInfo { Title = "ISM Sponsor API",  
Version = "v1" });  
options.SwaggerDoc("v2", new OpenApiInfo { Title = "ISM Sponsor API",  
Version = "v2" });
```

2. Request/Response Examples

Add more detailed examples using `[SwaggerRequestExample]` and `[SwaggerResponseExample]` attributes.

3. OAuth 2.0 Flow Integration

Configure Swagger UI to handle OAuth 2.0 authentication flows directly:

```
options.AddSecurityDefinition("oauth2", new OpenApiSecurityScheme  
{  
    Type = SecuritySchemeType.OAuth2,  
    Flows = new OpenApiOAuthFlows  
    {  
        AuthorizationCode = new OpenApiOAuthFlow  
        {  
            AuthorizationUrl = new  
Uri("https://login.microsoftonline.com/.../oauth2/v2.0/authorize"),  
            TokenUrl = new  
Uri("https://login.microsoftonline.com/.../oauth2/v2.0/token")  
        }  
    }  
});
```

4. Swagger Themes

Add dark mode and custom themes for Swagger UI:

```
options.InjectStylesheet("/swagger-ui/custom.css");
```

Testing Checklist

Development Testing

- ☐ Start application locally: `dotnet run`
- ☐ Access Swagger UI: <https://localhost:5001/api/docs>
- ☐ Verify all 25+ endpoints are listed
- ☐ Check that XML documentation appears for each endpoint
- ☐ Test "Authorize" button with a valid JWT token
- ☐ Execute a GET request (e.g., [/api/v1/reference/sponsors](#))
- ☐ Execute a POST request (e.g., [/api/v1/coverage/evaluate](#))
- ☐ Verify request/response schemas are accurate
- ☐ Download OpenAPI JSON: <https://localhost:5001/swagger/v1/swagger.json>

Pilot Testing

- ☐ Deploy to Pilot environment
- ☐ Access Swagger UI: <https://ismsponsor-pilot.azurewebsites.net/api/docs>
- ☐ Verify all endpoints accessible
- ☐ Test with Pilot environment credentials
- ☐ Validate cross-system integration endpoints
- ☐ Share Swagger URL with integration partners

Production Verification

- ☐ Deploy to Production
- ☐ Verify Swagger UI is **not accessible**: <https://ismsponsor.azurewebsites.net/api/docs> returns 404
- ☐ Confirm OpenAPI JSON is also disabled in Production
- ☐ Provide external partners with exported OpenAPI spec (from Pilot)

Related Documentation

- [API Contract Plan](#) - Complete API contract specification
- [Deployment Hardening Summary](#) - Production deployment guide
- [PowerSchool Integration Summary](#) - PowerSchool API details

Contact

For questions about Swagger implementation or API documentation:

- **Development Team:** support@ismsponsor.edu
- **Integration Support:** integrations@ismsponsor.edu